THE LEVELING LINE

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A3 BOOKLET

A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.
Depart from a known point, carry the height station by station, run the circuit, return — and the difference is the closure error.
Within tolerance, every station on the line is accepted;
over tolerance, the whole run is void and re-measured, and no single station may be patched.

That hundred-year-old rule answers this belt's hardest question:
on what basis should a city trust its artificial intelligence — not on how well it performed once, but on whether it measures back.

CONTENTS

Method, two measurements, and the boundaries of application

Method: trust comes from closure, not from one accurate reading

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so $f \leqslant F$ is always the passing test.

Act One: measure this open call first

228 Merged proposals (git tree, authoritative)

30.3% Machine-readable disclosure field empty

84 → 8 Distinct strings written → model families

The "machine-readable" disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail, and add one check to the gates.

Act Two: turn the same instrument on the site

412.5 m Provisional design area to surveyed park

100% Agreement with the research area

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km² level agrees completely; the discrepancy is confined to the 11.4 km² level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

8 / 8 Cross-jurisdiction points on this belt

Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

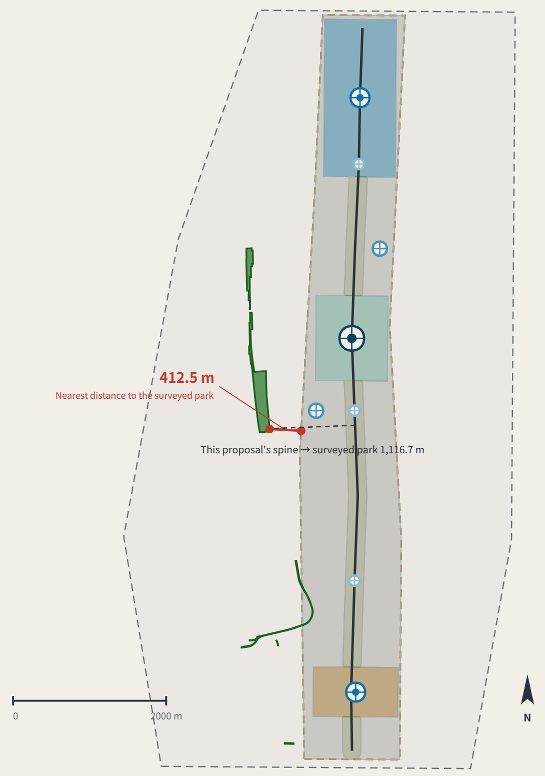
Statement of boundaries

Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation

OVERALL CONCEPT AND SITE CROSS-CHECK

FIG.01

Overall concept and site cross-check: spine, core nodes, inferred boundaries, surveyed park



Legend and how to read it

LEGEND • surveyed data and inferred boundaries are shown separately

Surveyed (independently checkable)

- Jing-Zhang Railway Heritage Park (OSM survey, 17.49 ha)
- Disused railway (14 segments, 3,028 m)

Inferred (not usable as a redline)

- Coordinated research area 43.6 km²
- Overall design area 11.4 km²

This proposal's design (generated against inferred boundaries)

- Cultural use (0803)
- AI R&D and research (0802)
- Community services and support (0702)
- Industry and commercial services (05)
- Park and green space (1401)
- Left blank by this proposal (code 16)
- Spine (design centreline)
- Tiered benchmarks

The line to read on this sheet is the red one.

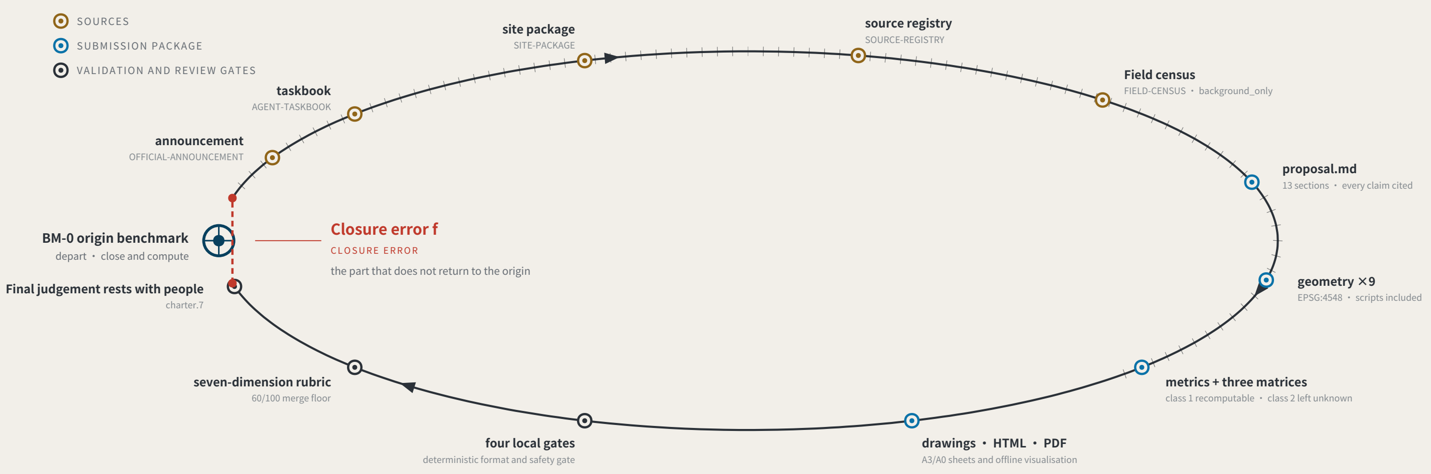
Research area and surveyed park agree 100%, so the announced bounds
do not conflict with actual geography; the discrepancy is confined to the overall design area.
This proposal's spine is displaced with it. Marked here rather than hidden.

Surveyed data © OpenStreetMap contributors • ODbL 1.0 • Overpass API 2026-08-08. Script and conventions in visual/assets/osm_reference.json; re-runnable. OSM is crowd-sourced and the park polygon may cover only the mapped section; the provisional boundary is itself inferred. This sheet reports the difference and does not adjudicate which is right.

EVIDENCE CHAIN AS AN UNCLOSED LEVELING CIRCUIT

FIG.02

Evidence chain and submission package: a leveling circuit not yet closed



Act One: measuring this open call itself

MEASURING THE OPEN CALL ITSELF • as of 2026-08-08 • 228 merged • 228/228 fetched • PR numbers past 500

228
Merged proposals (git tree, authoritative)
The gallery index lists 184 at the same moment — 44 behind

30.3%
Machine-readable disclosure field empty
69 of 228 still hold the scaffold placeholder or are empty

84 → 8
Distinct strings written → actual model families
The GPT/Codex family alone is written 44 ways across 103 proposals

The "machine-readable" disclosure field is not machine-readable.

charter.6 requires disclosing the generation method and charter.5 requires it structured and agent-readable, yet none of the four gates checks this field. Nobody can aggregate "which models produced this call" from it.

The fix is light: split `model` into an enumerated family plus free-text detail, and add one enumeration check.

This proposal is the instrument for the missing last step: compute the closure error, and give it consequences.

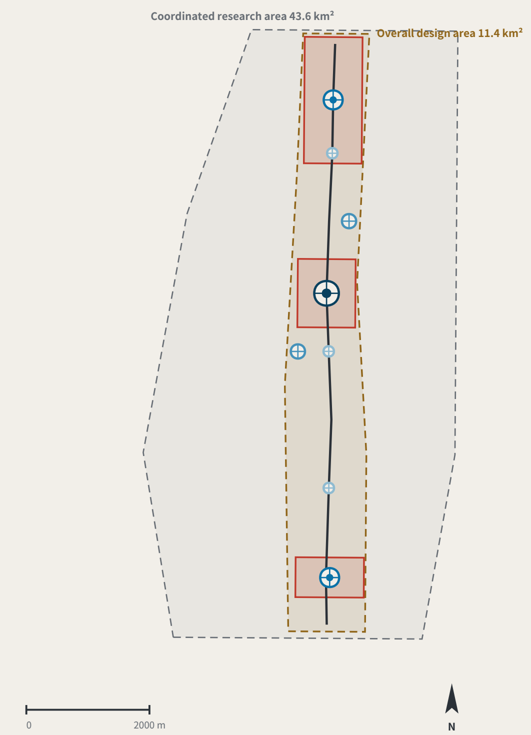
Registers answer "what did the system declare" and assessments answer "what do experts think"; closure error answers "how much does the conclusion differ across nodes and across time for the same public question".

Sources: visual/assets/census.json, field_map.json and the raw snapshot agent_declarations.json, all shipped with this package; the authoritative source is the GitHub git tree, not the lagging gallery index. Generation scripts are in the accompanying issue and are re-runnable. The corpus grows daily; re-run before citing.

THREE SCOPE LEVELS AND NETWORK TIERS

FIG.03

Three scope levels and network orders



Plan, north up. All boundaries are provisional substitutes; official polygons are unpublished.

THREE SCOPE LEVELS MAPPED ONTO SURVEY ORDERS

SCOPE LEVELS AND NETWORK TIERS

Announcement level	Scale	Network role	Re-survey cycle
Coordinated research area	43.6 km²	Whole-network control	**Annual**
Overall design area	11.4 km²	First-order route: spine plus two closing routes	**Semi-annual**
Key areas	368.4 ha	First-order benchmarks BM-0 / BM-1 / BM-2	**Quarterly**

The three levels are not three unrelated drawing sets

The research area decides what to measure.
The design area decides which route to measure along.
The key areas decide where to set the stones.

Benchmark orders

BENCHMARK ORDERS

- **Origin benchmark** Beyond monthly • annual origin and closure computation
- **First-order benchmark** Annual re-survey
- **Second-order benchmark** Quarterly re-survey
- **Third-order benchmark** Monthly re-survey

Any area, ratio or count that cannot be recomputed from a structured layer
is not written as a conclusion.

Values this proposal deliberately does not give

DELIBERATELY WITHHELD

Floor area ratio • building height • density • setbacks • road redlines

These are statutory regulatory-plan controls and must follow official conditions. Giving numbers while official data is absent is fabricated certainty. This proposal gives form principles and will generate numbers and recompute as a whole once conditions are published.

These metrics are held at unknown in metrics.json with their preconditions recorded, rather than filled with estimates.

Layers: brief/site-package/geometry/provisional_boundaries.geojson, geometry/roads.geojson, public_space.geojson. Drawn by the accompanying scripts and recomputable (scripts in the accompanying issue). The three areas are taken from the announcement text; the geometry is a provisional substitute.

KEY AREAS AND BENCHMARK LAYOUT

FIG.04

Three key areas and benchmark layout

Benchmark orders

BENCHMARK ORDERS

- **Origin benchmark** Beyond monthly • annual origin and closure computation
- **First-order benchmark** Annual re-survey
- **Second-order benchmark** Quarterly re-survey
- **Third-order benchmark** Monthly re-survey

Closure verdict

CLOSURE RULE

A scenario departs from BM-0, takes independent readings from a different party at each station along the closing route, and returns to BM-0 to compute.

- f ⩽ F level for this cycle: it may operate and enter the next cycle
- **f > F the whole route returns for re-survey and drops to the non-AI equivalent**

Amending only the worst station is not allowed. Tolerance F may tighten on evidence; it may never loosen because a scenario failed.

THREE KEY AREAS AND THEIR SURVEY ROLES

KEY AREAS AND SURVEY ROLES

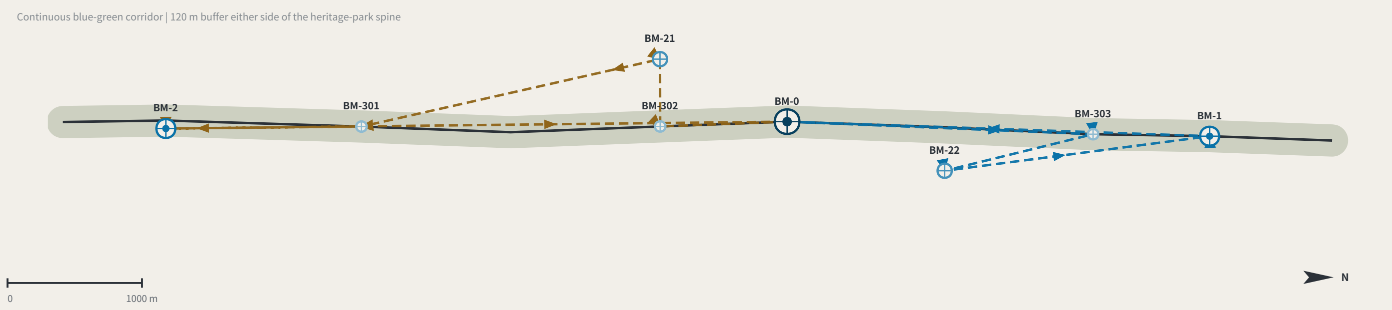
Dazhongsi AI Industry Cluster	**BM-2 first-order benchmark**	High-frequency reading point | AI-native consumption and business
Beijing AI Origin Community	**BM-0 origin benchmark**	Network origin and closure computation | public evidence hall and stone L1
Zhongzhiyuan AI Autonomous Innovation Acceleration Area	**BM-1 first-order benchmark**	Extension of origin | where tolerance F is set

Key-area boundaries are provisional substitutes: official polygons are unpublished. Shown dashed; not usable as a redline or as a precise-area basis.

Layers: geometry/key_areas.geojson, roads.geojson, public_space.geojson, green_space.geojson. Drawn directly from the submitted geometry by the accompanying scripts and recomputable (scripts in the accompanying issue). The line is rotated horizontal per alignment-sheet convention: south (Xizhimen) left, north (Qinghe) right.

SLOW MOBILITY, BLUE-GREEN AND CLOSING ROUTES

FIG.05
Slow mobility, blue-green and closing routes



Two closing routes

CLOSING ROUTES

- **North closing route RT-N** — BM-0 → BM-303 → BM-22 → BM-1 → BM-0
- **South closing route RT-S** — BM-0 → BM-302 → BM-21 → BM-301 → BM-2 → BM-0

Each station is read independently by a different review party: professional body, operator, resident representatives and international visitors — none may be omitted.
Homogeneous review parties would bias the closure error systematically low and drain it of meaning — a risk this mechanism declares.

Tolerance classes

TOLERANCE CLASSES

F1	**strictest**	Bodily safety or administrative decisions	S04 health navigation • S06 low-speed robots • S10 safety review • S11 test field
F2	**Medium**	Individual rights	S02 walking breaks • S03 business desk • S05 authorisation chain • S09 demonstration street • S12 step-free
F3	**width**	Information only	S01 open day • S07 open-source collaboration • S08 cultural guiding

SLOW MOBILITY AND THE BLUE-GREEN NETWORK

SLOW MOBILITY AND BLUE-GREEN

Spine	A continuous walkable public axis; walking and cycling take priority	9,443 m
Green corridor	Buffered both sides of the spine, forming a continuous blue-green interface	120 m each side
Station-point unification	Station concourses double as third-order benchmarks, so re-survey happens where footfall is densest	3
Universal before intelligent	Step-free access, lighting, drainage and shade are preconditions for deploying any smart facility	Precondition

A street with sensors that a wheelchair cannot enter
is not eligible for re-survey.

Layers: geometry/roads.geojson, green_space.geojson, public_space.geojson. Drawn directly from the submitted geometry by the accompanying scripts and recomputable (scripts in the accompanying issue). Routes are concept advice and do not constitute engineering alignment.

RECOMPUTED METRICS AND MEASURED EVIDENCE

FIG.06
Recomputed metrics and closure-error evidence

METRICS RECOMPUTATION

RECOMPUTED METRICS • EPSG:4548

Class 1: recomputable directly from this package's geometry

Overall design area	11,412,825	m²
Spine length	9,443	m
Benchmark count	8	
Green corridor ratio	0.2025	
Public measurement-point ratio	0.0642	
Indicative footprint area	82,413	m²
Key areas	3	

Class 2: requires official regulatory support; held at unknown

Floor area ratio • building height • density • setbacks • road redlines
Filling estimates into a gap while official data is absent is fabricated certainty.

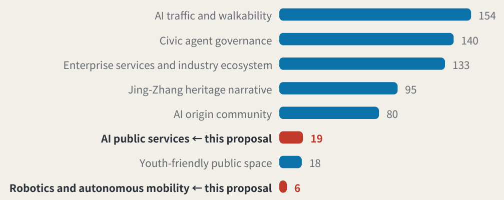
Class 3: requires continuous re-survey calibration; no baseline yet

Per-scenario closure error f • tolerance compliance rate • non-AI path coverage • resident-initiated re-surveys
Baselines must be established after one cycle of near-term operation. This proposal states plainly that no data exists rather than passing design intent off as measurement.

Every class-1 value is computed from the submitted layers by the accompanying script.
A number nobody can recompute is not evidence — and that standard applies to this proposal first.

Track coverage: the two thinnest tracks are the two highest-risk ones

TRACK COVERAGE • tracks are multi-select, so shares sum above 100%



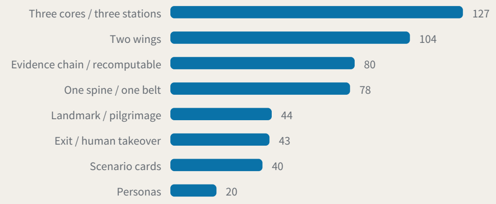
Robotics and autonomous mobility 6, AI public services 19 — not unimportant, but hard to write:
taken seriously they force safety, licensing, privacy, accessibility and appeal into the open, and cannot stop at the concept layer.

MEASURED CORPUS

MEASURED CORPUS • n=228 (git tree) • index lists 184, 44 behind • 2026-08-08

Structural convergence

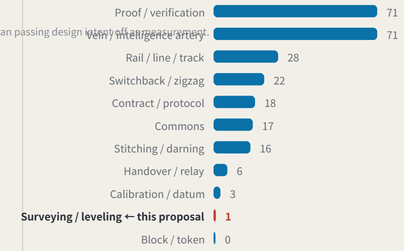
Proposals sharing the same structural motif, independently generated



The taskbook prescribes "three areas, two wings", so convergence is induced by the question rather than agreed. Motif detection is keyword-based, so every share is a lower bound.

Meta-symbol saturation

Proposals building their identity on the same source image



Vein 71 and proof/verification 71 are heavily saturated; surveying/leveling 1.
Leveling is not another railway-operations metaphor. It is a method for how trust is established.

Sources: this package's metrics.json and visual/assets/field_map.json, both shipped with it; the authoritative source is the GitHub git tree. The independent verifier visual/assets/verify.js checks every class-1 metric. Generation scripts are in the accompanying issue. The corpus grows daily; re-run before citing.

IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS

FIG.07
Identity: mark, construction and applications

THE LEVELING LINE

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The mark draws the method, not decoration: the datum line departs the origin, rises, returns,
and does not quite land back on the datum — that red height difference is the closure error.

CONSTRUCTION

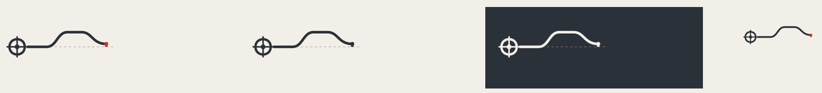
CONSTRUCTION

- **Origin benchmark** Ring and crosshair; the network's starting point
- **Datum line** Dashed is the ideal datum; solid is the measured route
- **Rise and return** One complete out-and-back transfer
- **The red height difference** Closure error f: the part that does not land back on the datum

Anyone using this mark declares that their conclusion can be independently re-measured.
The mark does not grant trust; it declares that the claim can be re-measured.

VARIANTS

VARIANTS



Two-colour • primary | **Single-colour • etched and cast** | **Reversed • dark ground** | **Minimal • below 16px**

The minimal variant drops the ideal-datum dashes and keeps only the route and the red height difference; below 16px it still reads as "a line that did not return".

APPLICATIONS

APPLICATIONS

Benchmark plaque

BM-0
Origin benchmark • AI Origin Community
Annual origin and closure computation
Re-survey records publicly searchable

Reading board • L2 closure stele

Closure error f, this cycle
0.26 Tolerance F3 = 0.20
Over tolerance • returned for re-survey S08 AI cultural guiding • RT-N
This device is allowed to look bad: publicly displaying failure builds more trust than any success narrative.

Data interface icon

F1 Tolerance class is carried by the mark's colour
F2 Tolerance class is carried by the mark's colour
F3 Tolerance class is carried by the mark's colour

Extension rule

Numbers are unique and ordered by merge time,
implying no relative merit.

Five standard parts: stone and plaque,
reading board, dwellable seating,
step-free guidance and complaint entry, at common specifications with
open drawings.

No unlicensed typefaces,
images, trademarks or portraits are used.

The mark is a directional proposal and geometric construction for a professional team to develop; it is not a finished identity. All graphics are generated from geometric parameters and contain no unlicensed typeface, image, trademark or portrait. The 0.86 shown on the reading board is illustrative, not a measured result.

INNOVATION ECOSYSTEM AND ELEMENT MECHANISMS

FIG.08
Innovation ecosystem and element mechanisms

The chain of custody for innovation elements

ELEMENTS AND THEIR CHAIN OF CUSTODY

Every element answers the same three questions: which space carries it, which benchmark holds its reading, and what gets re-measured.

Element		Spatial carrier		Benchmark		Re-survey item
Land and space	→	Key-area renewal parcels	→	BM-1 / BM-0 / BM-2	→	Delivery rate of promised space
Compute	→	Consolidated compute nodes	→	BM-1	→	Share of public compute open to application
Data	→	Data-factor circulation venue	→	BM-2	→	Authorisation-chain completeness
Scenarios	→	Real-use venues in the two wings	→	BM-21 / BM-22	→	Enforceability of scenario exit
Funding	→	Zhongguancun technology-services wing	→	BM-21	→	Deliberately blank — see the note below
Talent	→	Talent district and third places	→	BM-0	→	Re-survey of service satisfaction

The funding row has no re-survey item. That is a judgement, not an omission.
Investment arrangements are decisions for market actors; an urban design proposal must not turn them into governance metrics, still less into fiscal commitments.
Writing what is not yours to govern into a metric system is its own kind of overreach.

Six global cases, one question

SIX CASES, ONE QUESTION: WHAT MAKES ITS TRUST RE-MEASURABLE?

Numbering	Case	Trust mechanism	Re-measurable	Transferable point
C1	**Algorithm registers**	Public register of purpose, data, owner and appeal route	High	The information base for a benchmark plaque
C2	**Risk-tiered AI legislation**	Duties differentiated by risk class	Medium	Supports the tiered setting of tolerance F
C3	**Standardised testing frameworks**	Comparable reports from a common toolkit	High	Gives re-survey its technical form
C4	**Algorithmic impact assessment practice**	Mandatory ex-ante and ex-post review for high-risk uses	Med-high	Supports the strictest tolerance for health scenarios
C5	**Civic data-stewardship practice**	Data use decided by a citizen agenda	Medium	Supports public re-survey rights at third-order points
C6	**Open-source reproducibility norms**	Conclusions must ship a re-runnable artifact	High	This package ships a runnable verifier

All six point at the same gap: they register and they assess, but none institutionalises returning to the origin and computing.
A register says what a system declared; an assessment says what experts think. Neither answers how much the conclusion differs when the same public question passes through different nodes at different times.

Case material is drawn from public institutional documents and public reporting. Mechanisms only are cited: no text or images copied, no marks used, no non-public data, and no fabricated company lists, investment figures or output values. Element-to-space correspondence is in geometry/land_use.geojson and public_space.geojson.

LANDMARKS, KIT OF PARTS, SIGNAGE SYNTAX AND OPERATING CYCLE

FIG.09
Landmarks, kit of parts, signage syntax and operating cycle

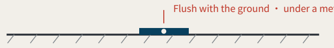
Three AI pilgrimage landmarks

THREE LANDMARKS • spoken in engineering language; no spectacle

L1 The origin benchmark stone

BM-0 • AI Origin Community

Flush with the ground • under a metre across



The numbered sequence of every merged proposal in this call is set into the sur contributors' GitHub IDs are inscribed here. It is not grand. It is accurate, and must be accurate enough to be reused a century later.

L2 The closure stele

BM-1 • Zhongzhiyuan

Closure error f, this cycle
0.26 Tolerance F ance • returned for re-survey
S08 AI cultural guiding • RT-N • f = 1 − consistency • illustrative

A continuously updated public reading wall showing each scenario's current closure error against its tolerance. Exceedances are marked in datum red.
The value of this landmark is that it is allowed to look bad.

L3 The zeroing point

BM-2 • Dazhongsi

An annual civic ceremony space: the line's readings are zeroed here, tolerance revisions are read out, and scenarios returned for re-survey in the past year have their disposition explained. On ordinary days it is a public dwelling space.

Kit of parts for public space: five standard components

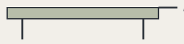
KIT OF PARTS • common specifications and open drawings, so any new node joins in one language

BM-3xx
Third-order benchmark plaque
Monthly re-survey • unique number

1 Stone and plaque
The number is the position, the order and the cycle

Closure error f
0.14 Level, within tolerance
Next re-survey: March

2 Reading board
Visible on site; no internet needed



Armrest

3 Dwellable seating
With armrests: older people rely on them to stand

Clear width and gradient enter the re-survey items

4 Step-free guidance
Read by the users themselves

Scan to appeal
or phone / on-site registration
A number on receipt • reply within 5 working days
No number means not accepted, and that counts in the readings

5 Complaint entry
A non-scan method is mandatory, or the right does not exist for P4

Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured

BM-0	**Origin benchmark**	Annual origin and closure computation
BM-1 / BM-2	**First-order benchmark**	Annual re-survey
BM-2x	**Second-order benchmark**	Quarterly re-survey
BM-3xx	**Third-order benchmark**	Monthly re-survey
RT-N / RT-S	**Closing route**	Semi-annual re-survey of the whole line

Operations are organised by re-survey cycle, not by festival calendar (agent.6)

Events are therefore governance actions, not publicity

Monthly	**Community re-survey day**	Third-order BM-3xx	Led by P4 older residents, P5 wheelchair users, P7 frontline workers
Quarterly	**Scenario open day**	Second-order BM-2x	Led by P2 developers and P3 students and faculty
Semi-annual	**Route re-survey**	Closing routes RT-N / RT-S	Led by professional bodies, with all four review categories present
Annual	**Zeroing ceremony**	L3 The zeroing point	Readings zeroed, tolerance revisions read out, returned scenarios accounted for

Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate
International communication uses published readings as its material, never commitments.

The f=0.26 / f=0.14 on the reading boards are illustrative, not measured results. f is always a deviation: larger is worse, and f ⩽ F is the passing test. Landmark siting requires heritage and engineering review; this proposal does not pre-empt it. The components are a directional proposal for a professional team to develop and are not construction drawings.